

# TUTORIUM 5:

P5.1:  $A \in \mathbb{R}^{n \times n}$ ,  $h: \mathbb{R}^n \rightarrow \mathbb{R}$ ,  $x \mapsto x^T A x =: h(x)$

Z.E.:  $Dh(x): \mathbb{R}^n \rightarrow \mathbb{R}^n$

$$f(x+s) = f(x) + Dh(x)[s] + o(\|s\|) \quad (s \rightarrow 0), \quad x, s \in \mathbb{R}^n \quad \text{"Klein Störung"}$$

$$\rightarrow h(x+s) \stackrel{!}{=} h(x) + Dh(x)[s] + o(\|s\|)$$

$$\begin{aligned} h(x+s) &= (x+s)^T A (x+s) = (x^T + s^T) A (x+s) \\ &= x^T A x + x^T A s + s^T A x + s^T A s \\ &= h(x) + x^T (A + A^T) s + o(\|s\|) \end{aligned}$$

verl.:  $Dh(x)[s] = x^T (A + A^T) s$   
 $Dh(x) = x^T (A + A^T) \in \mathbb{R}^{1 \times n}$

$$\lim_{s \rightarrow 0} \frac{o(\|s\|)}{\|s\|} = 0$$

$$\lim_{s \rightarrow 0} \frac{\|s^T A s\|}{\|s\|} = 0$$

$$s^T A x = (x^T A^T s)^T = x^T A^T s$$

$$Dh(x)[y] = \nabla h(x)^T y$$

$$\nabla h(x) = (\nabla h(x))^T$$

## Exkurs:

Def: DUALRAUM: Sei  $V$  ein <sup>reeller</sup> Vektorraum. Dualraum  $V^*$

$$V^* = \mathcal{L}(V, \mathbb{R}) = \{ \phi: V \rightarrow \mathbb{R} \mid \phi \text{ linear} \}$$

Satz: (Darstellungssatz von Riesz)

Auf endl. d. reellen Vektorraum  $(V, \langle \cdot, \cdot \rangle)$  gilt: Sei  $\phi \in V^*$ , dann:  $\exists! v_\phi \in V$  Riesz-Rep.

$$\phi(\omega) = \langle v_\phi, \omega \rangle, \quad \forall \omega \in V.$$

$$\bullet V \cong V^*$$

$\rightarrow \phi$  lebt im Dualraum  $\mapsto v_\phi$  im Vektorraum lebt.

Abb.

$$\Phi: V \rightarrow V^*, \quad v \mapsto \langle v, \cdot \rangle$$

$$\forall y \in \mathbb{R}^n: Dh(x)[y] = \langle \nabla h(x), y \rangle = (\nabla h(x))^T y$$

Riesz'sche Rep.

$$\langle x, y \rangle = x^T y$$

$$J_h(x) = x^T(A + A^T)$$

$$(J_h(x))^T = \nabla h(x) = (A^T + A) x$$

2. Stetigkeit

→ beschränkt

$$\text{Bew: } |I[f]| = \left| \int_0^1 f(x) dx \right| \leq \int_0^1 |f(x)| dx \leq \int_0^1 \|f\|_\infty dx = 1 \|f\|_\infty$$

$I$  ist (sogar Lip) stetig.

□

$$b) q: V \rightarrow V, q(f) = f^2$$

$$\text{z.z. } D_q(f): V \rightarrow V, D_q(f)(g) = 2fg$$

$$\text{Bew: } f^2 \Leftrightarrow f(x)^2 = f^2(x) \quad | \quad 2fg \Leftrightarrow (2fg)(x) = 2f(x)g(x)$$

$$\Delta \in V,$$

$$q[f + \Delta] - q[f] - D_q(f)(\Delta) = (f + \Delta)^2 - f^2 - 2f\Delta = \Delta^2$$

$$\Delta^2 \stackrel{?}{=} o(\|\Delta\|_\infty) \quad \lim_{\Delta \rightarrow 0} \frac{\|\Delta^2\|_\infty}{\|\Delta\|_\infty} = \lim_{\Delta \rightarrow 0} \|\Delta\|_\infty = 0 = o(\|\Delta\|_\infty)$$

→ Stetigkeit

1. linear

$$D_q(f)(\alpha\Delta + \beta\Gamma) = 2f(\alpha\Delta + \beta\Gamma) = \alpha D_q(f)(\Delta) + \beta D_q(f)(\Gamma)$$

$$2. \text{ Beschränkt: } \|D_q(f)(\Delta)\|_\infty = \|2f\Delta\|_\infty \leq 2\|f\|_\infty \|\Delta\|_\infty \Rightarrow \text{stetig}$$

$$\hookrightarrow F = I \circ q$$

$$DI[q] = I, \forall q \in V$$

$$DF(f)(\Delta) = D(I \circ q)(f)(\Delta) = (DI[q](f)) \circ D_q(f)(\Delta) = I \circ (2f \Delta) = 2 \int_0^1 f(x) \Delta(x) dx$$

P5.3:  $V = (C([0,1], \mathbb{R}), \|\cdot\|_\infty)$

$$f: [0,1] \rightarrow \mathbb{R}$$

Sup Norm:  $M$  ein nicht leere Menge  
( $X, \|\cdot\|_X$ ) Norm. Raum:

$$\|\cdot\|_\infty: B(M, X) \rightarrow \mathbb{R}, \|f\|_\infty = \sup \{ \|f(x)\|_X \mid x \in M \}$$

$$V^* \quad F: V \rightarrow \mathbb{R}, F(f) = \int_0^1 [f(x)]^2 dx$$

(a) z.z.:  $I: V \rightarrow \mathbb{R}, f \mapsto \int_0^1 f(x) dx$  linear + stetig.

Bew: 1. Linear:  $f, g \in V, \alpha, \beta \in \mathbb{R}$

$$I[\alpha f + \beta g] = \int_0^1 (\alpha f(x) + \beta g(x)) dx = \alpha \int_0^1 f(x) dx + \beta \int_0^1 g(x) dx = \alpha I[f] + \beta I[g]$$

P5.2:  $\Phi: \mathbb{R}^+ \times \mathbb{R} \rightarrow \mathbb{R}^2$ ,  $\Phi(r, \varphi) = \begin{pmatrix} r \cos \varphi \\ r \sin \varphi \end{pmatrix}$

(a) Existenz + Stetigkeit  $\forall \partial_i \iff$  Total Diffbar.  $\checkmark$

$$\Phi(r, \varphi) = \begin{pmatrix} \Phi_1(r, \varphi) \\ \Phi_2(r, \varphi) \end{pmatrix} \Rightarrow \Phi \text{ stetig!}$$

$$J_{\Phi}(r, \varphi) = \begin{pmatrix} \partial_r \Phi_1 & \partial_{\varphi} \Phi_1 \\ \partial_r \Phi_2 & \partial_{\varphi} \Phi_2 \end{pmatrix} = \begin{pmatrix} \cos \varphi & -r \sin \varphi \\ \sin \varphi & r \cos \varphi \end{pmatrix}$$

(b)  $\nu_r: \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}^2$ ,  $\nu_r(\Phi(r, \varphi)) = \begin{pmatrix} \cos \varphi \\ \sin \varphi \end{pmatrix}$

$\nu_{\varphi}: \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}^2$ ,  $\nu_{\varphi}(\Phi(r, \varphi)) = \begin{pmatrix} -r \sin \varphi \\ r \cos \varphi \end{pmatrix}$

$$J_{\Phi}(r, \varphi) = (\nu_r \quad \nu_{\varphi})$$

Z.Z: Wohldif. +  $\langle \nu_r, \nu_{\varphi} \rangle = \langle \nu_{\varphi}, \nu_r \rangle = 0$

$$\Phi(\tilde{r}, \tilde{\varphi}) = \Phi(r, \varphi + 2\pi k) \quad \forall k \in \mathbb{Z}$$

$$\tilde{r} = r, \quad \tilde{\varphi} = \varphi + 2\pi k$$

$$\begin{aligned} \bullet \nu_r(\Phi(\tilde{r}, \tilde{\varphi})) &= \nu_r(\Phi(r, \varphi + 2\pi k)) = \begin{pmatrix} \cos(\varphi + 2\pi k) \\ \sin(\varphi + 2\pi k) \end{pmatrix} = \begin{pmatrix} \cos(\varphi) \\ \sin(\varphi) \end{pmatrix} = \nu_r(\Phi(r, \varphi)) \\ \bullet \nu_{\varphi}(\Phi(\tilde{r}, \tilde{\varphi})) &= \nu_{\varphi}(\Phi(r, \varphi + 2\pi k)) \\ &= \begin{pmatrix} -r \sin(\varphi + 2\pi k) \\ r \cos(\varphi + 2\pi k) \end{pmatrix} = \begin{pmatrix} -r \sin(\varphi) \\ r \cos(\varphi) \end{pmatrix} = \nu_{\varphi}(\Phi(r, \varphi)) \end{aligned}$$

$$J_{\Phi}^T(r, \varphi) = \begin{pmatrix} \nu_r \\ \nu_{\varphi} \end{pmatrix}$$

$$\begin{aligned} J_{\Phi}^T(r, \varphi) J_{\Phi}(r, \varphi) &= \begin{pmatrix} \nu_r \\ \nu_{\varphi} \end{pmatrix} (\nu_r \quad \nu_{\varphi}) = \begin{pmatrix} \langle \nu_r, \nu_r \rangle & \langle \nu_r, \nu_{\varphi} \rangle \\ \langle \nu_{\varphi}, \nu_r \rangle & \langle \nu_{\varphi}, \nu_{\varphi} \rangle \end{pmatrix} \\ &= \begin{pmatrix} \cos \varphi & \sin \varphi \\ -r \sin \varphi & r \cos \varphi \end{pmatrix} \begin{pmatrix} \cos \varphi & -r \sin \varphi \\ \sin \varphi & r \cos \varphi \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} \end{aligned}$$

$$F = \left( \int_0^1 \int_0^{2\pi} f(x) dx \right)$$